ActInf Textbook Group ~ Cohort 2 ~ Meeting 9 (Chapter 4, part 2)

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Hello, everyone. Thanks for joining. It's October 28th, 2022.

ActInv Textbook Group Cohort 2. We're on the 9th.

Oh, we're in the...

The weeks are shifted back by one. Is that true?

There's two 21s.

Or there's been some kind of other time glitch.

But today is 1028.

Anyways.

We're discussing Chapter 4.

Continuing our discussion of Chapter 4.

So, where would anyone like to start with?

Does anyone have a general Chapter 4 remark?

Or a particular question that they want to look at?

Or anything else?

Daniel, it's Bronwyn.

Mm-hmm.

There's just a lot in Chapter 4, obviously.

And it sort of...

I think it takes in a lot of different aspects of active inference

and then sort of piles them all in there

and then throws out a whole lot of maths.

So, you know, it's unbelievably dense.

But...

And, you know, I don't know any maths.

But interestingly, as I read it,

there's some sort of themes that come out

that are sort of simplistic.

And I'm wondering if somehow that can be sort of...

If you can sort of summarise that today,

if that makes sense.

It's just sort of like a...

A capsulated paragraph of the process it goes through.

That was one thing.

The other thing was...

The Box 4.1, the Message Passing and Inference.

I didn't quite get the difference
between Variational Message Passing
and the Belief Propagation.
Whether it can talk a bit about that.
It's about...
Yeah.
Yeah, that's about a bit early.
Okay.
Does anyone have any...
Thought on this in chat?
Or just anything else to add?
Or we can...
Try to address these.
Okay.
Okay.
Well, one way to look at the theme
of Chapter 4
or the kind of
looking beyond all of the
technical details.
And the technical details,
it's kind of like an iceberg
because they could show a little
or they could show a ton.
Every time they show one,
they could also show a whole
set of ways to get to it or not.
So it's kind of like...
Maybe...
It's like a variable of the text.
How much of the math they just mention
versus explicitly write out.
Yeah.
Because there's a lot of papers
with no equations at all,
like a philosophy paper,
that will still say,
because of the variational free energy

or because the generative model
is this way.
So kind of referencing
the same topic.
And so...
But different texts differ
in how much formalism they have.
What is the role of the generative model
in active inference?
What does anyone think about that?
Or in the textbook,
where is any quote
where they use...
Where they say,
a generative model is blank.
Or this is the role of the generative model.
We can look for some too.
Or any other thought someone has.
I mean,
I think it says that they can vary
depending on the inference problem.
That's one thing about generative models.
Okay.
Yeah.
Another way to...
That variation point's really important.
The variation is sometimes described
with like fine-tuning
within a given inference problem,
which is sometimes framed as perception
in the most fast timescale
and learning and memory
in a medium timescale.
Yeah.
And then also
different implementations
or different like model structures
can vary across inference problems.

There's also different ways
to like write
and see the generative model.
Can anyone just like list
or suggest
how are generative models
shown or represented?
So is that the factor graphs?
Yeah.
The 4.3.
In which section?
Figure 4.3.
Is it...
4.2.
So how...
Yeah.
One view
of the generative model
is like this
figure 4.3
kind of
classic slash common representation.
The Bayes graph view
with the nodes
being a random variable
and edges
being relationships
among variables.
And so like
if every variable
were connected
to every variable
like
every time point
were causal
on every time point
every hidden factor
influence

every hidden factor
you just have
like a fully connected
all by all
causal
association matrix.
so that's like
the weakest
world model.
And then
you can
say well
the present
is like
the only
intermediating
Markov blanket
between the future
and the past.
Or
this is the
you know
perception
is the only
intermediating
Markov blanket
between observations
and hidden states.
If you are
willing to take
those
structural
assumptions
and embody
them in the model
the sparsity
of the model
under

many different
ways of calculating
the model
it will
it just
like
always gets easier.
You might be
restricting yourself
to a very local
neighborhood of
models
but pretty much
the more
you can constrain
the edges
and the combinatorics
people can
add other
remarks on this
simpler models
are just
simply easier
to fit
better
optimization
is always
achieved
by making
quadratic
or curved
type
assumptions
it's like
a general
optimization
principle
to make

things
on a
smoother
more
incrementally
updatable
related
problem
they deal
with time
differently
they deal
with everything
differently
time
slash
space
etc
differently
sorry
can you
hear me
now
yes
Ali
go ahead
yeah
I just
wanted to
mention
this
paper
a tale
of
densities
which
clears up
some of

the
misunderstandings
or misconceptions
around
generative
model
because
more often
than not
people
think of
generative
model
as just
a
structural
representation
of the
of the
entire
of the
external
world
but
in active
inference
the
generative
model
is
usually
represented
in terms
of its
recognition
density
so
this

recognition
density
is
something
that
enables
us
to
or let's
put it
this way
it's an
approximate
posterior
probability
density
that inverts
generative
mapping
from
consequences
to
causes
so
it enables
us
to
recognize
the hidden
states
based on
the
observations
and
inferring
and
inferring
the

causes
of
sensory
outcomes
and not
just
simply
as
a
static
representation
of the
external
world
great
point
Ali
to
like
kind
of
unpack
this
and
this
is a
great
entry
point
the
tale
of two
cities
is a
book
about
two
cities

the
tale
of
two
densities
is some
work
with
Ramsted
and
others
and
there's
at least
some
stream
on it
and
the
two
densities
are like
the
two
directions
which
a
Bayesian
type
model
can be
used
which
is
in
its
recognition
capacity

and
in
its
generative
capacity
so
one
point
is the
difference
between
a
sort
of
like
descriptive
statistical
approach
whether
with
p
values
or
any
other
type
of
statistics
where
you're
just
taking
observables
only
and
describing
them
statistically

or
summarizing
them
the
two
densities
are
those
two
bases
of
the
Bayesian
model
which
is
about
its
capacity
to
generate
data
from
hidden
states
and
as
Ali
said
to
be
inverted
or
to
be
able
to
perform

inference
on
hidden
states
from
observables
!
and
that is
exactly
the
distinction
of
S
and O
S
are
unobserved
hidden
states
of the
world
which
in a
causal
graphical
framework
there's
different
interpretations
of what
it
means
that
causal
force
but
it's

the

GM

family

that

one

is

embodying

and

then

the

observations

are

up

so

blue

I'm

sorry

I

just

was

going to

add to

that

if

you're

done

always

love

listening

to

that

live

stream

and

Maxwell

stated

it

nicely
the
differences
between
the
recognition
density
and the
generative
density
it was
how he
described
it
was
if
you're
stacking
up
dominoes
and
then
you're
going to
knock
them
over
so
the
recognition
density
is
the
stack
of
dominoes
and
the

generative
density
is
like
the
action
occurring
between
the
dominoes
falling
down
like
I
always
like
that
and
it's
locked
into
my
mind
and
really
kind
of
helps
me
get
a
grip
on
maybe
the
differences
there
okay

um
Brock
or
someone
what
have

you
added
this
quote
for
um
just
on
page
64
this
is
like
a
referencing
equation
4.1
which is
uh
right here
it just
says
that
um
that
the
generative
model
is
the

point
that
prior
and
likelihood
are
sufficient
to
compute
all
evidence
and
probability
so
I guess
it's
a
slightly
more
formal
specific
definition
and
narrow
because
right
after
that
it
talks
about
um

computational
tractability
of doing
that
and

that
seems
like
the best
starting
point
of
the
generative
model
is
blank
in the
book
that I
was
able
to
find
nice
good
summary
like
in the
frog
jumping
example
prior
and
likelihood
left
side
prior
prior
beliefs
on
likelihood
of the

hidden
state
on the
probability
of the
hidden
state
before
the
observation
likelihood
model
the
larger
matrix
it has
a different
dimensionality
than
this
one
the
prior
and
it's
the
mapping
of
different
things
then
there's
an
observation
but
in a
sense
none

of
these
beliefs
or
likelihood
models
are
observed
the
observation
o
which is
like the
experimental
data
actually
being
collected
is here
and then
everything
else are
like the
cognitive
states
or internal
states
or states
as modeled
by the
modeler
and then
there's
some
updating
and in
chapter
two

that was
being
described
in terms
of
the
surprise
self-information
bayesian
surprise
but in the
context of
what
barack just
read
prior and
likelihood
comprise a
generative
model
prior
likelihood
that already
compose a
generative
model
but it's
like a
car that
hasn't been
run
and then
there's some
that's when
data starts
being able to
update the
model

so that's
why many
times
we see
equations
like a
bayesian
equation
whether
represented
graphically
or with
like
in like
just a
single line
form
because
to specify
a bayesian
statistical
problem
is to
have specified
the generative
model
and vice
versa
Ali
and I
think it's
a general
issue
regarding
the whole
active
inference
literature

because
whenever
we read
a paper
in active
inference
we need
to be
quite clear
about
what do
the authors
mean by
each
concept
because
for instance
in this
paper
from
2022
learning
generative
models
for active
inference
using
tensor
networks
they have
defined
generative
model
in its
traditional
statistical
form
and not

exactly
it doesn't
exactly
maps
onto
this
somehow
more
comprehensive
or modified
definition
of generative
model
in terms
of recognition
density
and so
on
so
yeah
and it
also extends
to every
other concept
we encounter
in any other
active
inference
paper
thanks
great
point
um
yep
that's
good
point
the

purpose of
just bringing
back to
chapter
two
and this
example
is just
to
um
situate
everything
that's
happening
with the
theoretical
like data
free
math
of act
inf
and also
the data
oriented
aspect
of act
inf
which is
uh
to give
a deflationary
perspective
a specified
bayesian
statistical
model
and then
data

that that
model has
legibility
around
of course
you can
you know
have
you know
it could be
the wrong
model for
the situation
and so
on but
this is like
this is like
the statistical
scenario
which is
just prior
likelihood
data
and then
this posterior
can become
that one's
prior
and that's
just a
uh
yeah
can be
modeled
different ways
in continuous
time
in discrete

time
and again
the whole
point is
just
reduce the
uh
theoretical
novelty
of active
inference
because
uh
it's
more
like
bayesian
statistical
modeling
in this
mode
than
many
many
other
things
and so
a lot
of the
questions
and it's
it's
there
there are
great
questions
that come
up and

have come
up in
statistics
before but
kind of in
a new
way here
like
these are
general
statistical
questions
they're
statistical
questions
involving
action
building
on
some of
the
causal
statistical
developments
from
pearl
and others
who
brought
in
like
just
the
concept
of
the
markov
boundary

in
in that
like
90s
80s
90s
and
beyond
period
um
it's
using
those
kinds
of
statistical
techniques
developed
for
Bayesian
statistical
settings
in the
last
five to
50 years
in the
setting of
action
and
niche
dynamics
also
connecting
in some
way that's
not fully
characterized

yet
with
some
more
physics
like
formalizations
like the
Bayesian
mechanics
but that's
not the
first kind
of connection
between
statistical
optimization
and
physics
theories
which is
kind of
the whole
idea of
like why
you can
kind of
roll the
ball
to the
bottom
of the
hill
physically
and then
the
statistical
optimization

works like
that too
so
the
generative
model
chapter
to kind
of return
to
four
and
the
first
part of
the
Bronwyn's
earlier
question
I think
is this
chapter
is
describing
many
important
aspects
and
two
fundamental
types
of
generative
models
those
dealing
with
active

inference
in
a
discrete
time
formalization
and a
continuous
time
formalization
because there's
also other
ways to do
it
probably
it's
just giving
technical
details
on what
those
generative
models
are
because
that is
the
entity
description
that's
the unit
of
perception
cognition
action
that's
the
empirical

model
as
used
in the
computer
to
recognize
data
to generate
data
so
this is
like
kind
of
the
core
idea
and
sometimes
it's
dealt
and
like
yeah
sometimes
it's
just
mentioned
but
when
it's
mentioned
that's
why
this
is
the

textbook
so
that
in
some
other
philosophy
paper
they're
not
going
to
have
this
but
this
is
kind
of
what
is
being
discussed
when
those
terms
are
used
in
qualitative
settings
or
in
conversation
with
Ali's
caveat
of course

that
everyone
is
using
it
differently
and
then
in
terms
of
like
scientific
review
and
in
evaluating
like
a
technology
or
some
claim
or
something
if
someone
says
the
generative
model
does
this
or
it
has
this
characteristic

these
are
like
the
specific
aspects
that
would
characterize
that
claim
being
true
so
that was
good
very
interesting
discussion
on
the
generative
model
how
about
the
box
Ali
well
as far
as I
understand
that
the
difference
between
variational
message

passing
and
belief
propagation
basically
comes
to
the
difference
in a
holistic
view
of
the
approximate
inference
computation
and
the
recursive
or
step-by-step
computation
of it
so
in
variational
message
passing
case
we
actually
have
the
already
actualized
trajectory
for

messages
and we
know
all the
conditional
properties
of
children
with regard
to their
parents
and with
those
informations
we can
compute
the whole
approximate
free energy
of the
whole process
but
for
belief
propagation
we need
to compute
that
recursively
in each
step
without
necessarily
needing
to have
all the
information
for all

the
conditional
probabilities
of
parents
for
I'm
sorry
children
with regard
to parents
beforehand
so
but
another
important
difference
between
them
is that
the
belief
propagation
takes
account
for
all
the
messages
that
has
happened
up to
the
point
of
B
or

the
point
that
we're
momentarily
dealing
with that
point
so
because
of
that
we
divide
the
whole
process
with
the
probability
of
B
because
we
don't
want
to
include
B
in
the
process
yet
so
that's
basically
something
similar

to
a
kind
of
deductive
way
of
doing
inference
and
the
inductive
way
of
doing
that
okay
awesome
thank
you
Ali
that's
very
insightful
let's
look at
4.1
and see
where we're
seeing
some
aspects
of
this
well
let's
pull
out

one
more
level
Ali
what
would
you
say
what
category
of
things
do
variational
message
passing
and belief
propagation
what is
the
category
that
even
makes
these
things
comparable
well
the
similarities
between
at least
the
structural
similarity
between
these
two

ways
of
computation
is
their
dependency
on
the
expectation
of the
conditional
probability
so
in each
case
we're
just
computing
the
inference
in
terms
of
their
expectation
and
as
we
see
even
the
equations
look
quite
similar
to
each
other

so
but
as
they
pointed
out
in
the
bottom
of
the
box
this
is
a
slightly
non-standard
use
of
the
expectation
operator
so
we
need
to
be
aware
of
this
point
as
well
because
well
if
both
of

those
expectation
operators
were
exactly
the
same
there
wouldn't
be
any
difference
any
similarities
between
those
two
kinds
of
inference
schemes
all
right
thanks
a lot
for
that
a few
notes
one is
these
will be
fun
to
see
the
descriptions
of

and
just
yeah
like
what
these
expectations
are over
keeping
in mind
the expectation
here is
about
the
central
tendency
of
a
statistical
distribution
not
a
time
projection
so
talking
about
as
Ali
highlighted
the
expectations
on
variables
that
doesn't
necessarily
have

to be
a
time
series
prediction
in
fact
one
of
the
most
important
uses
is
just
to
calculate
a
rolling
snapshot
in the
context
of
prior
and
likelihood
distributions
as
incoming
data
come
in
so
they're
like
parameter
bidding
modeling

approaches
and
is it
fair to
say
that
for
a
given
analytically
expressed
Bayesian
equation
so
formula
it's
possible
to
take
the
approach
of
working
with
it
more
along
one
branch
or
more
along
another
branch
is that
the case
Ali
or anyone

else
papers
tend to
like
take
one
approach
or
another
but
there's
also
some
papers
to
probably
compare
them
but
they're
distinct
implementations
potentially
with
some
differences
in
outcome
that would
be
interesting
to
explore
but
is it
the case
that
they're

both
possible
to be
used
for a
given
Bayesian
equation
if we
wanted
to do
the frog
jumping
out of
the
hands
could
there
be
a
variational
message
passing
and
a
belief
propagation
implementation
well
I guess
so
if
the
kind
of
situation
we're
dealing

with
doesn't
necessarily
restrict
us
to
use
the
global
view
versus
the
local
view
of
the
inference
then
I guess
both of
them
can
be
applied
let's
save
this
way
sorry
just
yeah
the
variation
message
passing
is
kind
of

let's
say
omniscient
like
view
of
the
situation
because
we
know
all
the
conditional
probabilities
of
each
node
but
for
the
belief
propagation
we're
dealing
with
the
explore
exploit
dilemma
also
their
computational
implementations
could be
different
so
a given

language
or package
might
only
enable
one
and
one
other
well
let's
return
in the
last
20
minutes
back
to
the
text
and
if
we
go
there
we
go
there
but
this
also
relates
to
the
factor
graphs
and
some

mathematical
equivalencies
with
factor
graphs
and
Bayesian
graphs
that
is
some
of
the
work
since
like
2017
and a
little
bit
before
of
DeVries
and
Par
and
Friston
where
these
algorithms
can
statistically
have
good
computational
implementations
or
just

how
good
exactly
probably
is an
open
question
but
these
but
there
are
procedural
rules
for
calculating
large
graphs
of
this
structure
which
is why
they
represent
recent
developments
in
Bayesian
computational
statistics
it would
be
awesome
to hear
from
someone
who's

working
with
those
kinds
of
large
scale
implementations
but
that
is
my
kind
of
outside
interpretation
and
also
there
are
some
recent
developments
in
the
for
making
these
kinds
of
computations
more
tractable
and
reducing
the
complexity
classes

of
the
computations
one
example
is
this
recent
algorithm
called
branching
time
active
inference
developed
by
and
others
and
also
the
work
done
by
Barron
Milledge
such
as
the
inference
learning
or
the
hybrid
predictive
coding
is also
an attempt

to reduce
the complexity
classes
of these
kinds
of
implementations
which
can be
I believe
can be
a promising
path
towards
achieving
to
a more
tractable
and
computationally
effective
way of
computing
these
inferences
nice
thank you
for that
yeah
there's
sort of
like
a
meme
like
machine
learning
is

statistics
and
all
that
so
these
this
chapter
and
hopefully
seeing
how
some
of
these
developments
from
pure
Bayesian
statistics
but
in
the
more
recent
era
there
have
also
been
computational
kind
of
interplay
with
the
pure
theory

development
for
example
bootstrap
based
algorithms
non-parametric
statistics
where one
like
resamples
from
distributions
and does
that many
many times
that kind
of sampling
based
approach
was
worked
out
on
paper
long
before
it became
computationally
plausible
to actually
sample
and store
large amounts
of information
and then
once that
computational

availability
started to
happen
and better
sampling
algorithms
and better
hardware
then
that area
of statistics
can become
applied
which
sometimes
requires more
theoretical
development
but often
not
it's often
just simply
a question
of the
implementation
being
written
from like
a code
or
computational
implementation
perspective
so
there's
a lot
of
equivalences

in
act
inf
but I
think
the
juxtaposition
of
figure
4-3
generative
models
with
Markov
blankets
first
and
then
saying
these
are
different
heuristics
computational
implementations
model
fitting
approaches
and
others
others
that
if you
can state
it like
it looks
in figure
4.3

then
this
kind of
computational
statistical
branch
will become
accessible
it's like
if they
showed
here's
the linear
regression
and
here's
the R
method
or here's
the
Python
script
that you
can use
or
here's
the
optimization
here's
three ways
that you
can optimize
a linear
regression
and there
are multiple
ways to do
linear

regression
and there's
different software
packages associated
with them
too
so it's
like if
your
GMs
have this
structure
then
these are
optimization
implementation
implementation
approaches
that very
importantly
are not
sampling
based
now
complexly
enough
sampling may
have to
come into
play
in the
real
computational
implementation
but that's
kind of a
deeper
or more

specific
question
to a
given
way that
it's
applied
but just
more
broadly
these
are not
just
trying to
sample
data
until the
landscape
is like
fully
coded
these
are
models
that
with their
sparsity
like what's
not connected
embody
sometimes
incredibly
strong
structural
assumptions
about the
world
which is

why there's
the whole
question of
structured
learning
because once
you're
structured
in the
GM
like we'll
see it in
chapter 6
as well
the kind
of structure
is set
and then
there's like
a fine
tuning
where the
actual
variables
get tuned
to the
right
values
that's
learning
and
perception
but also
learning can
be
structural
so I
hope that

kind of
provides
some
concordances
with the
generative
model
and speaks
to its
like
total
centrality
in
act
inf
in chapter
4
and connects
it to
box
4-1
that's
describing
some
exciting
techniques
for
computing
large
base
graph
type
generative
models
one
last comment
on this
and then

we'll still
have more
time
just time
for
kind of
overviewing
again
is like
I think
people have
certainly
said before
don't anyone
specific right
now but
just that
there is
a lot of
focus on
the Markov
blanket
concept
and the
sort of
like
everything
about the
Markov
blanket
the mystery
but less
focus on
the generative
model
concept
so when
people then

thinking
about
how do
we model
organizations
or
this
this
cognitive
feature
there's
a lot
of
focus
on
like
well
what
are
the
nested
Markov
blankets
and
now
there's
a way
to
think
about
that
as
nested
Markov
blankets
recognizing
the generative
model

that
isn't
just
the
entity
scale
blanket
but
that
kind of
nested
Markov
blanket
representation
multi-scale
systems
kind of
like
just
general
point
what
gets
often
not
brought up
in those
situations
is like
okay
society's
nested
Markov
blankets
but
what
are
the

generative
models
if
you
specify
the
generative
model
you'll
have
also
specified
the
Markov
blankets
because
you'll
have
defined
the
particular
things
embodied
in the
actual
generative
model
so
it's
like
is it
an
active
inference
claim
that
a
situation

has
been
analyzed
in
terms
of
interacting
generative
models
and
generative
processes
or
is it
a
general
claim
about
multi-scale
systems
being
nested
so
when we
think about
just the
encapsulating
matryushka
nature
of the
nested
blankets
degenerative
model
is not
specified
in more
cases

than not
hope
that's an
interesting
and fair
point
okay
we have
10 to
12 minutes
left
is there
any other
like
thought
or
question
Brock
wrote
spooky
inference
in close
proximity
to surprise
generates
noisy
definitions
i was
just
kidding
about
ali
saying
lots
of
different
names
for

things
we call
that
spooky
action
at a
distance
entanglement
now
but
at the
time
it was
a bunch
of
different
things
when it
was first
observed
and it
was
surprising
so
anyways
um
i don't
know
if
uh
yeah
it's
worth
like
uh
thinking
or just
linking

like
lingering a
little bit
on the
intractability
of
specifying
like
the
model
entirely
in most
in a lot
of cases
yeah
thanks
yeah
i think
that kind
of
um
anticipates
some
discussions
that are
in like
chapter
six
with the
recipe
for
how to
make
the
generative
model
and
the

difference
between
like
fully
specifying
a quote
toy
model
which is
to say
like a
model
where you
can
see it
graphically
at like
this
like just
kind of
like this
like this
is the
whole
model
um
in that
case
you may
be able
to
create
a little
information
bubble
where that
is the
full

model
specification
but then
in any
real
situation
even
like
doing
research
let alone
doing
anything
with
causal
feedback
in the
loop
the
fully
specified
model
just
is
somewhere
between
doesn't
exist
can't be
specified
won't be
specified
shouldn't
be specified
so then
what
what
what

then

Okay, any

thoughts on four or let's just look quickly to five?

Ali, go for it.

Yeah, just one small point here.

I'm not sure if I mentioned it before or not, but there's kind of another type of active

inference.

Of course, we have very different classes or categories of active inference, but one specific

type is called sophisticated inference, which kind of deals with the situations in which,

well, I mean, instead of predicting what would become later at each stage, it tries to compute

everything that has come before or in order to have a way to compute counterfactual consequences

or the things that we need to, or somehow the kind of decision-making processes and so on.

And interestingly, in the paper, Branching Time Active Inference, they've stated that this

type of active inference is a subset of Branching Time Active Inference.

I mean, if we just take that Branching Time Active Inference and separate it into forward

and backward propagation trajectories, we'll come with traditional active inference and sophisticated

active inference.

So I think the main paper for sophisticated active inference is this one that I'm posting here.

So that's just one small point I wanted to mention.

Great point.

Thanks for sharing this generative model knowledge.

It's very helpful.

Livestream 11 was sophisticated affective inference.

So that was with a secondary or, you know, here, verbally secondary adjective, affective, reflecting

that they were modeling an affective variable or taking an affective interpretation of their

model's cognitive function.

They were doing affective inference in the context of sophisticated inference, which is as Ali described.

Allows for a kind of mapping and modeling of past, present, and future, which is like why in 4.3,

there's like a time minus 1 and time plus 1, it could have been written as just time, time plus 1, time plus 2, and been like a two-step planning problem.

But also there's, you can choose different pieces of the time horizons to model. Ali?

Or a better way to, probably a better way to put it is to say that in situations where we have beliefs about beliefs,

this kind of sophisticated inference can come to play.

Instead of just beliefs about states.

And the composability of Bayesian graphs,

is high.

Which is what allows all of these different structural models to be explored.

Like nested models that represent spatial enclosure.

Nested models that represent counterfactuals.

Models that consider the past and are able to reconsider the past observations.

Models that don't.

All of those, like a memoryless process.

Wow.

All right.

Chapter five is quite different and we'll just look through it and see how different it is.

It's going to pick up on this message passing topic, which will be cool to revisit again and consider it primarily from the context of neurobiology.

In this chapter, we focus on the process theories that arise from chapter four.

Central to this implementation of Bayesian belief updating is the notion of Bayesian message passing.

Microcircuits and messages.

Cortical layers.

Part of the mammalian brain.

Mammal neuroanatomy.

Mammal computational cerebral neuroanatomy.

Abstract hierarchical predictive coding model.

Motor reflex arc.

In the spinal column of the mammal.

In the spinal column of the mammal.

In the spinal column of the mammal.

Policy selection.

Computational neuroanatomy in the mammal brain.

Putative roles of neurotransmitters in active inference.

Different neurotransmitters.

Different neurotransmitters.

And they're sort of like.

Putative.

Or.

Coarse grained or hypothesized.

Computational role.

And different studies that support that computational role.

If.

If.

If.

If.

If.

If.

If.

If.

If.

If.

If.

If.

If.

that computational role for that neurotransmitter. Not saying it's like the only thing it does or the only way to look at it, just they found empirical support for it playing that statistical role in those experiments. Then some early moves towards integrating continuous and discrete models, which returns later in the second half of the book. And the closing figure is showing all of the three examples from neuroanatomy together. Planning with those dopaminergic brain regions, then the prefrontal cortical regions on top, and the spinal reflex motor behavioral arc with action. So it's like when we've been talking about planning as inference and prediction and generalization categorization as inference, learning as inference, action as inference, this is the graph. So it's a different chapter. It's not as equation driven. And uh, it's primarily based on those examples of mammal and generalized computational neuroanatomy. So it's a cool last chapter for the first half. All right. Any other last comments?

I'll stop recording. Thank you.